phenylalanine and intake of ascorbic acid. It must be noted that dietary restriction may be promising for children but not for adults.

7.6 TAY-SACHS DISEASE

Gangliosides are glycosphingolipids which are main components of cell membranes. Nervous tissues are particularly rich in gangliosides. They constitute a significant fraction (6%) of brain lipids. Other tissues also contain gangliosides but in lesser amounts. Their complex carbohydrate head groups, which extend beyond the surfaces of cell membrane, act as specific receptors for certain glycoprotein hormones. They act as receptors for some bacterial toxins. Gangliosides are also involved in cell – cell recognition. Generally, the carbohydrate segments of glycolipids are removed bylysosomal hydrolases in the early phases of the turnover of these compounds. Disorders of ganglioside break down are responsible for several hereditary sphingolipid storage diseases including Tay – Sachs disease.

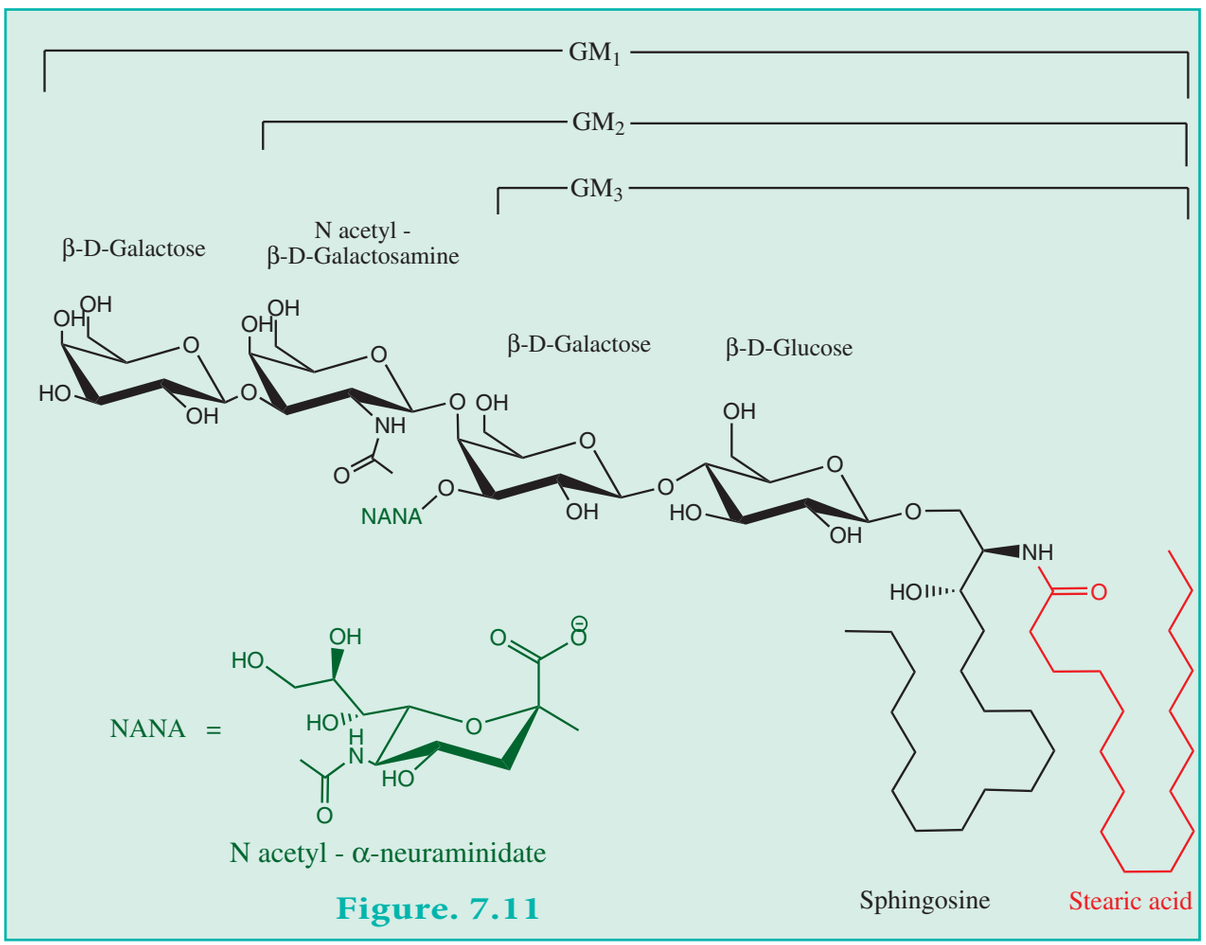


Figure. 7.11

β-galactosidase hexaminadase-A

GM1 → GM2 → GM3

Tay-Sachs Disease is an autosomal recessive neurodegenerative disorder. It is common in Jewish population which is fatal within the first three to four years of life. A British physician, Warren Tay, and an American physician, Bernard Sachs independently worked on this disease.

XII U7 Inborn.indd 179 4/8/2020 3:22:39 PM